
Is the Diminishing>Returns Nonlinearity of Color Perception Cone-Aligned?

An Identifiability-Aware Model Comparison on Public Data

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(automated analysis; code and logs accompany this report)

Abstract

Bujack et al. (2022) showed that suprathreshold color-difference perception exhibits *diminishing returns* (DR)—strict subadditivity along a geodesic—so perceptual color space cannot be Riemannian. Their 2025 follow-up found DR is not constant across color space and conjectured a compact explanation, a *per-cone decomposition* $D = f(f_L(\Delta L) + f_M(\Delta M) + f_S(\Delta S))$, noting it “cannot be tested on measurements from two geodesics only.” We test that conjecture on existing public data: four human dissimilarity datasets ($\sim 92k$ gray-axis 2AFC trials; 12k lightness triads; the Ekman and Helm hue circles) placed in a normalized CIE 2006 cone space and compared under a pre-registered protocol with cross-validation and selection-aware bootstrap inference—a nested family (additive M0, isotropic-DR M1, per-cone-DR M2) against a matched-complexity panel of per-axis DR models on five standard non-cone bases (opponent, sRGB, CIE RGB, CIELAB, CIE LCh). The gray-axis DR finding replicates in full (held-out gain of M1 over M0 507.0 nats; concavity ratio 0.767[0.752, 0.782] vs. 0.757 implied by the 2022 report). Against this anchor the cone-aligned hypothesis fails twice. First, per-axis structure helps little overall: M2 improves on a single global scaling by only 8.2[−4.9, 21.7] nats (not significant). Second, and decisively, the cone orientation is *worse* than standard non-cone alternatives: the selection-aware deficit of M2 against the best panel member (the CIE RGB axes) is −73.7[−132.4, −31.3] nats, and M2 also loses to the opponent basis (−54.7[−108.1, −7.1]). Synthetic discrimination analysis at the exact real designs shows this error direction does not arise under cone-aligned truth (rate 0.00 across 50 replicates; misalignment detected at ≥ 0.98 when real), so the deficit is evidence against cone alignment, not low power. Every basis that beats the cones contains a red–green opponent-like axis that the fits endow with strong compression—a compressive nonlinearity sited *after* receptor combination. An identifiability analysis (collinearity diagnostics, synthetic recovery, powered coarse-level comparisons) delimits what these data resolve: the finest per-cone split is weakly identified, and verdicts are stated at the supported granularity, before and after hue-locus curvature control. On the directions public data span, DR is channel-organized only weakly, and where it is organized, it is not organized by the cones.

1 Introduction

For over a century, perceptual color space has been modeled as a three-dimensional Riemannian manifold: the perceived difference between two colors is the length of the geodesic that connects them [Riemann, 1854, von Helmholtz, 1867, Schrödinger, 1920], and the CIE color-difference metrics are built on this premise. A Riemannian metric implies *additivity along geodesics*: for

A, B, C in order on a geodesic, $D(A, B) + D(B, C) = D(A, C)$. Bujack et al. [2022] demonstrated *diminishing returns* (DR), the strict subadditivity $D(A, B) + D(B, C) > D(A, C)$, on the neutral axis—overturning the Riemannian paradigm. Stark et al. [2025] then showed DR on two chromatic lines and, crucially, that its strength is *not constant*: a single global scaling $f(\Delta E)$ fit on one line fails on the other. This rules out the simplest non-Riemannian correction (one location-independent scaling) and raises the next structural question, which they pose explicitly but could not test: is the DR nonlinearity organized by the cone photoreceptor channels, i.e., does the metric take the form

$$D = f(f_L(\Delta L) + f_M(\Delta M) + f_S(\Delta S)), \quad (1)$$

“with four 1D scaling functions”? As they note, “this hypothesis cannot be tested on measurements from two geodesics only” [Stark et al., 2025]. Testing it requires dissimilarity data spanning *many directions* of cone space.

Why cone alignment matters. The question is not merely descriptive; its answer changes what can be predicted and what can be inferred about the visual system. (i) *Compact prediction.* If Eq. (1) holds, three per-cone functions plus one outer scaling generate DR *everywhere* in color space from a handful of measurements—a compact generative model, instead of the direction-by-direction atlas Stark et al. [2025] concluded would otherwise be needed. (ii) *Mechanism localization.* Alignment localizes where the compressive nonlinearity sits: aligned to the cones (L, M, S), compression applies to each channel *before* combination—at or near the receptor stage; aligned to post-receptor *derived* axes ($L+M, L-M, S-(L+M)$) [Derrington et al., 1984], it acts *after* aggregation—a different mechanism at a different site. Our coarse-granularity comparison (Sec. 3.2) makes this pre- vs. post-combination contrast empirical. (iii) *What a null would mean.* If the cone-aligned model is no better than standard non-cone bases, DR is neither cone-organized nor cleanly channel-localized on that axis system, and the organization must be sought elsewhere—its own informative, and why we test a *panel* of standard fixed bases rather than one control: matching any of them falsifies cone-specificity.

Throughout, mechanism claims are calibrated to what the data resolve: the headline is a *negative* verdict on cone alignment, with a weaker positive indication of opponent-like organization, and we quantify the discrimination power behind both.

What we do. We answer the cone-alignment question with existing, public human data only—no new data were collected. Four datasets span complementary directions of cone space: the Bujack et al. [2022] gray-axis 2AFC trials (91896 trials, 320 triads); the Wiebel et al. [2017] lightness MLDS triads (plain view, 12,000 trials); and two classic near-isoluminant hue circles with per-stimulus CIE coordinates: Ekman [1954] (14 monochromatic lights) and Helm [1964] (10 Munsell tiles). All stimuli enter a stated, normalized CIE 2006 cone space [Stockman and Sharpe, 2000, CIE, 2006] via a documented conversion with fixed gauge. We compare, with triad/pair-disjoint 10-fold cross-validation and selection-aware bootstrap intervals: **M0**, an additive weighted city-block baseline (no DR); **M1**, a single concave outer scaling (isotropic DR—the correction Stark et al., 2025 showed is insufficient); **M2**, per-cone concave inner scalings (Eq. 1); and **M3**, a required panel of matched-complexity controls with the same functional form on five standard *fixed, non-cone* bases (opponent; sRGB; CIE RGB; CIELAB; CIE LCh). The cone-alignment claim requires M2 to beat the *best-performing* member of this panel on held-out data, with the selection of the best control re-done inside every bootstrap resample.

Contributions. (1) The first direct test of the per-cone DR conjecture of Stark et al. [2025], on public data spanning many cone-space directions, under a pre-registered, selection-aware protocol (analysis_plan.md). (2) A replication anchor: the 2022 gray-axis DR finding reproduces qualitatively and in magnitude (held-out $M1-M0 = 507.0[309.0, 717.2]$ nats on the 2022 data; concavity ratio $0.767[0.752, 0.782]$ vs. the 2022-implied 0.757 ; independently confirmed on Wiebel et al., 2017, $74.0[37.1, 114.8]$ nats). (3) The headline verdicts: direction-dependence beyond one global scaling is weak on these data ($M2-M1 = 8.2[-4.9, 21.7]$ nats, n.s.); the cone orientation is *rejected* as the organizing basis—M2 underperforms the best non-cone control by $-73.7[-132.4, -31.3]$ nats (selection-aware), an error direction that never occurred under a cone-aligned ground truth (rate 0.00 over 50 replicates). (4) The identifiability analysis the conjecture has lacked: collinearity diagnostics, synthetic parameter recovery, and discrimination power against the nearest control—delimiting the granularity at which any channel claim can be made from public data—plus hue-locus curvature controls appropriate to each data type. (5) A calibrated mechanistic reading: every basis that beats

the cones contains a red–green opponent-like axis carrying strong compression—compression sited after receptor combination.

Restating that DR exists or varies across color space is Bujack et al. [2022] and Stark et al. [2025]’s result, not ours; mere anisotropy (M2 beating M1) would largely restate non-constancy. The novel object is *cone-alignment*—Eq. (1) against matched non-cone alternatives—and the identifiability limits of testing it on public data.

2 Related Work

The non-Riemannian program. Bujack et al. [2022] introduced the modern DR test: 2AFC triads on the neutral axis, analyzed with a Thurstonian maximum-likelihood model in which a perceptual map g (position along the axis) is estimated first and a difference-scaling function f second [building on Krantz, 1967, Maloney and Yang, 2003, Teti et al., 2022]; a concave f implies DR and hence non-Riemannian geometry. A controlled in-person study replicated the crowdsourced result [Stark et al., 2024]. Stark et al. [2025] (“Now in Color”) extended the paradigm to two chromatic lines at $L^*=70$ in CIELAB (green–pink and blue–orange), found concave f on both, and showed *non-constancy*: the scaling learned on one line loses predictive power on the other. Their Discussion poses the per-cone decomposition (Eq. 1) and states it cannot be tested from two geodesics. Bujack et al. [2025] formalize hue, saturation, and lightness in a non-Riemannian setting; that work is theoretical (with targeted experiments on shortest paths to the neutral axis), not a per-cone empirical test. Our study begins exactly where Stark et al. [2025] stops: we supply the many-direction test they call for on public data, plus the identifiability analysis that determines what it can resolve. We re-measure nothing they measured; our contribution is the alignment verdict, its controls, and its limits.

Older evidence on large color differences. DR at suprathreshold differences has a long history: MacAdam [1963] fit psychometric power laws with exponents below one; Helm [Helm, 1964] found distance estimates consistent with logarithmically compressed inter-point distance; Izmailov and Sokolov [1991]’s spherical model implies chord–arc subadditivity; Judd [1968] argued an “ideal” additive color space cannot exist. We reuse the classical Ekman [1954] and Helm [1964] data, but unlike the MDS tradition we never embed freely: stimulus coordinates are fixed colorimetrically and *metric hypotheses* are compared on them.

Channel structure in color vision. The cone fundamentals are standardized [Stockman and Sharpe, 2000, CIE, 2006, Stockman, 2019]; post-receptor opponent organization ($L-M$, $S-(L+M)$, $L+M$) is classical [Derrington et al., 1984]. Compressive transduction per cone is standard in color appearance modeling (cube-root-type responses in CIELAB and IPT-class spaces [CIE, 2004, Ebner and Fairchild, 1998]). What has been missing is a direct test of whether the *difference-scaling* nonlinearity (the DR effect, as distinct from the static transducer) decomposes along the cones or along derived axes; that is the gap this paper addresses.

3 Methodology

All modeling choices below were fixed in a time-stamped analysis plan (proposal/analysis_plan.md) written before any model was fit; deviations and corrections are disclosed in Sec. 4.4. Code and logs sufficient to re-derive every number are provided.

3.1 Working space: cone conversion, normalization, and transduction

Fundamentals and bridge. We use the provided CIE 2006 (Stockman–Sharpe) 10° LMS fundamentals (linss2_10e_1.csv, energy-linear, unit peak) and CIE 1931 2° CMFs (ciexyz31_1.csv), both from cvrl.org. The datasets specify stimuli in 1931 coordinates, so a 3×3 bridge M with $LMS \approx MXYZ$ is required; none is exact (2° vs 10° observers) and none was shipped, so we estimate M by least squares over the common wavelength grid and disclose its quality: per-cone R^2 0.9992/0.9992/0.9781 ($L/M/S$). The bridge is verified on known colors: the equal-energy white (matrix route vs. exact spectral sums: angular error 0.46°, worst component error 2.9%) and a chromatic reference on the spectral locus (600 nm: angular error 0.53°). The bridge is weakest exactly on the spectral locus (small- S region), where it can produce slightly negative S (at 537, 555 nm);

therefore the monochromatic Ekman stimuli are placed *directly* on the fundamentals at their wavelengths (exact by construction, as that dataset’s documentation prescribes), with the matrix route retained only as a logged cross-check (working-space angular deviations up to 17.8° , dominated by the near-zero- S stimuli). The Helm tiles (in-gamut surface colors) use the matrix.

Von Kries normalization (stated gauge of the space). Each dataset’s stimuli are expressed relative to that dataset’s adapting white at luminance $Y=100$: Illuminant C for Helm (its specification white), the equal-energy white for Ekman (spectral lights; the exact spectral EE white is used), and the respective neutral white for the two gray-axis sets—for which the normalized coordinates are *exactly* (t, t, t) with $t = Y/Y_n$, independent of the assumed white chromaticity and of M (verified numerically). Cross-dataset comparability of absolute DR strength therefore rides on this adaptation-relative convention, which we state rather than hide; the alignment verdicts are driven by within-dataset structure.

Static transducer. The working coordinates are $\tilde{c} = (\text{LMS}_{\text{vonKries}})^{1/3}$ component-wise. This fixed cube-root stage is the position map (the “perceptual scale” g of Bujack et al., 2022), adopted *by convention* because (i) the 2022 analysis found $\hat{g}$ on L^* near-linear—our replication (Sec. 4.3) refits $\hat{g}$ freely and confirms it (max deviation $8.0[7.3, 8.7]$ L^* units, the 2022 crispening bump); (ii) cube-root cone compression is the standard position stage of CIELAB/IPT-class spaces [CIE, 2004, Ebner and Fairchild, 1998], keeping the CIELAB control structurally comparable; (iii) it makes “difference scaling” well-posed off the gray axis with no free parameters. DR in this paper is always the *difference-scaling* nonlinearity on top of these fixed position coordinates—exactly the 2022 separation of g (position) from f (difference scaling). The transducer exponent is sensitivity-swept (Sec. 5.5).

3.2 Model family

For a stimulus pair with working-space difference $\Delta\tilde{c} \in \mathbb{R}^3$, every model has the form

$$D = f\left(\sum_{k=1}^3 w_k g(|z_k|/m_k; \beta_k); \alpha\right), \quad z_k = \langle u_k, \Delta\tilde{c} \rangle, \quad (2)$$

with inner scalings $g(x; \beta) = \log(1+\beta x)/\beta$ (so $g(x; 0) = x$) and outer scaling $f(u; \alpha) = \log(1+\alpha u)/\alpha$; $w_k \geq 0$, $\beta_k \geq 0$, $\alpha \geq 0$.

Gauge (fixed a priori, data-independent). The family in Eq. (2) is unidentifiable without a gauge: a global concave f with linear g ’s is observationally close to linear f with concave g ’s, and amplitude trades between f , the g ’s, and the links. We fix $f(0) = 0$, $f'(0) = 1$, $g(0; \beta) = 0$, $g'(0; \beta) = 1$; amplitude lives in the axis weights ($\sum_k w_k = 3$) and per-dataset link scales (Sec. 3.3); the outer f is restricted to the stated one-parameter concave log family, placing per-axis flexibility in (w_k, β_k) . The m_k are *design constants* (the median of the nonzero $|z_k|$ over the pooled stimulus-pair design, computed from stimuli only; the nonzero restriction avoids degeneracy for axes that vanish identically on the gray sets, e.g. Δa^*): this is the whitening that equates optimizer conditioning across bases. The family is exactly invariant to per-axis rescaling (absorbed by (w_k, β_k)), so no basis gains expressiveness from axis scaling. One known near-degeneracy remains and is quantified rather than hidden: on data spanning a single direction, inner (β) and outer (α) curvature trade off; they are separated only by multi-direction data (Sec. 5.3).

Models. **M0** (additive baseline, no DR): cone axes $u_k \in \{e_L, e_M, e_S\}$, all $\beta_k=0$, $\alpha=0$; a linear weighted city-block distance. It is an additive (L1) metric, not itself Riemannian (L2); the object of study is the DR nonlinearity within an additive-aggregation family, and the L1-vs-L2 aggregation choice is noted as a limitation (Sec. 6). **M1** (isotropic DR): cone axes, $\beta_k=0$, α free—one global concave scaling of the additive distance, the correction that Stark et al. [2025] showed is insufficient across two lines. **M2** (per-cone DR): cone axes, $\beta_L, \beta_M, \beta_S, \alpha$ free—the empirical form of Eq. (1). **M2c** (coarse cone, pre-combination): L and M channels tied ($w_L=w_M$, $\beta_L=\beta_M$), S separate. **M3 panel** (matched non-cone controls; all fixed, disclosed bases; none fitted): the same form and *identical* free-parameter count as M2 (six shape parameters) on (i) the opponent axes $\{(L+M), (L-M), (S-(L+M))\}$ (unit-normalized coefficient vectors in $\tilde{c}$ -space); (ii) the sRGB axes and (iii) the CIE RGB axes, i.e., the rows of the standard XYZ→RGB matrices [IEC, 1999, Wyszecki and Stiles, 1982] pushed to cone-space directions $u_k \propto (A_k M^{-1})^\top$ and unit-normalized;

(iv) CIELAB and (v) CIE LCh, which are perceptually defined nonlinear spaces and therefore enter through their *own fixed coordinate differences* ($|\Delta L^*|, |\Delta a^*|, |\Delta b^*|$) and ($|\Delta L^*|, |\Delta C_{ab}^*|, |\Delta H_{ab}^*|$) (standard formulas, dataset reference white, fully disclosed; no per-stimulus or tuned linearization). Each control has exactly M2’s complexity and the same whitening rule; a fitted free rotation is excluded from the panel because it nests M2 and would invert the test. **M3c-opp** (coarse post-combination control): opponent basis with the two chromatic axes tied—matched to M2c (four shape parameters). The coarse pre- vs. post-combination question is M2c vs. M3c-opp: $g(|\Delta \tilde{L}|) + g(|\Delta \tilde{M}|)$ (compress, then combine) against $g(|\Delta(\tilde{L} + \tilde{M})|)$ (combine, then compress).

Nesting: $M0 \subset M1 \subset M2$ exactly (verified numerically in the logs), and $M2c \subset M2$, $M3c\text{-opp} \subset M3\text{-opp}$. M2c does *not* nest M1 (it ties the weights), so the M2c-vs-M1 comparison is non-nested; the analysis plan’s shorthand “ $M0 \subset M1 \subset M2c \subset M2$ ” was corrected on this point during the audit, with the fitted models and comparisons unchanged (Sec. 4.4). As a supplement—explicitly *not* part of the matched M2-vs-M3 test—we also fit a free-orientation variant (M2 on a fitted rotation $R \in SO(3)$ of the cone axes) and report it as an orientation-recovery analysis (Sec. 5.6).

3.3 Per-dataset observation models

The datasets sit on different response scales and are never pooled into one naive likelihood. A single shared distance function D (one set of shape parameters w, β, α) is linked to each dataset by its own observation model:

$$\text{Bujack 2AFC: } P(R=1) = \Phi(s_B [D(s, t_2) - D(s, t_1)]) \quad (\text{binomial over 320 triads}), \quad (3)$$

$$\text{Wiebel triads: } P(R=1) = \Phi(s_{W,\text{subj}} [D(s_2, s_3) - D(s_1, s_2)]) \quad (\text{binomial; 10 per-observer scales}), \quad (4)$$

$$\text{Ekman ratings: } d = c_E (1 - e^{-a_E D}) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_E^2) \quad (\text{bounded Shepard-type link}), \quad (5)$$

$$\text{Helm ratings: } d = a_H D + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_H^2) \quad (\text{ratio-scale estimates; linear link}), \quad (6)$$

with each Helm series pre-normalized by its own mean (fixed preprocessing). The 2AFC likelihoods follow the reference implementation of Bujack et al. [2022] (σ absorbed into s). Two structural facts are stated up front. First, for rating datasets a scalar monotone link is confounded with the *outer* f ; DR-*presence* claims therefore rest on the link-pinned forced-choice datasets, while rating datasets contribute per-axis (*inner*, direction-dependent) structure, invariant to any scalar monotone link. The saturating Ekman link (its $d = 1 - \bar{s}$ ratings compress near 1; Shepard, 1987) and the linear Helm link (ratio-scaled distance estimates) are the pre-registered primaries; both alternatives are run as sensitivity specs. Second, the datasets barely overlap in direction (gray sets: achromatic ray; hue circles: near-isoluminant shell), so cross-dataset identification of shared curvature is expected to be weak—part of the identifiability story, not a nuisance to hide.

3.4 Fitting, cross-validation, and comparison protocol

Likelihoods are maximized by L-BFGS-B on log/softmax-transformed parameters (constraints by construction), 12 seeded multistarts per fit; every fit logs seed, per-start objectives, and convergence flags (1.000 of best fits converged; multistart agreement within 0.5 nats for most starts). Folds are fixed once (seed 20260816) before any fitting: 10-fold CV with the fold unit a *stimulus configuration*—triad-disjoint for the triad datasets (Bujack stratified by standard; Wiebel folds shared across observers), pair-disjoint for the hue circles. Test folds never influence model selection or any hyperparameter; per-observer nuisances are estimated on training folds only. The primary metric is total held-out log-likelihood (nats) summed over the ten test folds, reported jointly and per dataset block.

Uncertainty and selection-awareness. Held-out differences carry 95% intervals from a stratified cluster bootstrap ($B=1000$) over held-out units (triads/pairs, within dataset), recomputing every model’s total per resample. Because the headline control is *selected* as the best panel member, the M2-vs-best-M3 interval re-identifies the argmax control *inside each resample* (equivalently, the interval of the paired $\min_k [M2 - M3_k]$); a naive interval around the point-selected best control would be

winner’s-curse-inflated and is not used. Parameter and curve uncertainty use $B=200$ refit bootstraps on the full data (trial-level within triads for the 2AFC sets, pair-level for Ekman, observer-level for Helm), labeled as descriptive rather than held-out inference. M2’s extra parameters are never treated as evidence: comparisons are held-out at matched folds (full-data AIC/BIC, appendix, agree with the held-out ordering).

3.5 Identifiability analysis

Because ΔL and ΔM are strongly correlated over the stimulus designs and the datasets span few direction families, the finest per-cone split may not be identifiable. We quantify this with: (1) *design diagnostics* computed from stimuli only (correlations of per-axis difference magnitudes, singular values of the cone-vs-opponent rotation set, and the fraction of direction variance along the $L-M$ subspace where the cone-vs-opponent rotation lives); (2) *synthetic parameter recovery*, clearly labeled synthetic: responses simulated at the exact real designs (same triads/pairs/counts, fitted nuisance scales) from full-data fits of M1, M2, M2c, the opponent control, and the CIE RGB control, then refit; (3) *discrimination power*: the full cross-validated comparison is re-run on every replicate and rates of significant verdicts tabulated—power to detect cone alignment when true, false-positive rate when a control is true, and, critical for our observed outcome, the probability of a control *significantly beating* M2 under cone-aligned truth. The opponent basis is designated a priori as the nearest control (its rotation from the cones lies mostly in the weakly sampled $\Delta L \approx \Delta M$ subspace), so discrimination against it is the binding requirement; the CIE RGB generator was added after the primary results identified it as the best-performing control (disclosed as a post-hoc addition; the pre-registered generators are unchanged and reported). (4) *powered coarse comparison*: M2c vs. M1 and M2c vs. M3c-opp with the same protocol, so channel organization at the coarse level is itself controlled against a matched non-cone alternative, with power estimated from the same synthetic suite.

3.6 Hue-locus curvature confound control

Subadditivity along a *curved* stimulus locus arises even under an additive metric (chord vs. arc), so curvature control is logically prior to any chromatic DR claim. *Triad data*: the gray-axis stimuli are exactly collinear in the working space (the neutral locus is a straight ray; verified numerically at $\leq 1.5e-06^\circ$), so the near-collinear-triple restriction is satisfied exactly and curvature cannot mimic DR there. *Pairwise hue circles*: we control curvature two ways. (a) An explicit arc-vs-chord comparison against an additive-metric baseline: observed path-sum ratios $\rho(i, s) = \sum \text{adjacent } d / d_{\text{direct}}$ are compared with the same ratios from the fitted additive chord model M0 (curvature-only, through the same fitted link); “before control” reads $\rho > 1$ naively as DR, “after” attributes only the *excess* over M0 geometry to DR. (b) A near-adjacent restriction: the model comparison re-run using only pairs within ≤ 3 steps (Ekman) and ≤ 2 steps (Helm) along the locus, where the locus is close to straight (chord/arc 0.982 and 0.898 respectively, vs. 0.695 for Helm antipodes). Residual aliasing that no reanalysis of pairwise data can remove (curvature of the *perceptual* embedding for far pairs) is acknowledged in Sec. 6.

4 Experimental Setup

4.1 Datasets, licenses, preprocessing

Bujack 2022 gray-axis 2AFC [Bujack et al., 2022]: 91896 crowdsourced trials over 320 triads ($L_{t_1}^* \leq L_s^* \leq L_{t_2}^*$) with standards $L_s^* \in \{30, \dots, 70\}$ and test offsets up to 30 in steps of 2.5; response $R=1$ iff the (s, t_2) pair was judged more different (coding verified against the authors’ reference likelihood and by response monotonicity). From the `lanl/color` repository (BSD-3). The release carries no participant identifiers, so folds are triad-disjoint rather than participant-disjoint (disclosed). **Wiebel 2017 lightness MLDS** [Wiebel et al., 2017]: method-of-triads “which pair is more different,” 10 observers $\times$ 120 triads $\times$ 10 repeats in the plain-view context (12,000 trials); the four transparency contexts are excluded because they engage lightness constancy that our stimulus model does not represent (disclosed restriction). Reflectance $\rightarrow$ luminance uses the deterministic mapping in the released matching files (maximum 312.2 cd/m², used as the anchoring white; Gilchrist et al., 1999). License CC-BY 4.0 (OSF). **Ekman 1954 hue circle** [Ekman, 1954]: mean similarities among 14

monochromatic lights (434–674 nm) from ~ 31 observers, converted to dissimilarities $d = 1 - \bar{s}$ (91 pairs); redistributed machine-readably via the `smacof` package [de Leeuw and Mair, 2009] (GPL-3); equal-brightness aperture colors placed directly on the provided fundamentals (Sec. 3.1). **Helm 1964 hue circle** [Helm, 1964]: per-observer ratio-scaled interpoint distance estimates for 10 Munsell Value-6 tiles (45 pairs), 11 color-normal series (N1–N10 plus the N6 repeat) used, 4 color-deficient observers excluded (disclosed; cone-normal question); colorimetry from Helm’s Table I under Illuminant C; also redistributed via `smacof` (GPL-3). The provided cone-fundamental tables are from `cvrl.org` and are cited rather than relicensed [Stockman and Sharpe, 2000, CIE, 2006]. Converted working-space coordinates for every stimulus are logged (`logs/stimuli_*.csv`); the four datasets contribute 116/44/91/45 distinct stimulus pairs and 320/1200/91/495 grouped likelihood rows. Figure 6 shows the stimulus geometry in the working space.

4.2 Folds, bootstraps, and held-out likelihood

Folds: 10-fold, fixed once from seed 20260816 with the units of Sec. 3.4 (320 Bujack triads, 120 Wiebel triads, 91 Ekman pairs, 45 Helm pairs); joint fold k = union of per-dataset folds k ; assignments logged (`logs/cv_folds.json`). Held-out likelihood is the sum over folds of test-fold log-likelihood under that fold’s training-fit parameters and training-fold nuisances. Cluster bootstrap: $B=1000$ over held-out units (stratified by dataset; seed logged); refit bootstrap for parameters: $B=200$. Every quantitative claim in this paper traces to a log file produced by the run scripts; the numbers are typeset from machine-generated macros.

4.3 Gray-axis replication summary

Before any new claim, the pipeline must reproduce the 2022 finding. We reimplemented the two-stage 2022 analysis exactly (monotone-spline $\hat{g}$ on L^* , then $\hat{f}$ in the four published families; $\sigma=1$; 10 splits with 50 held-out responses/triad; 100 bootstrap training sets of 48,000 trials). Results (Fig. 1): held-out response-prediction accuracy orders *baseline* (0.8860 ± 0.0010) $< g$ *only* (0.9067 ± 0.0004) $< g+f$ (spline 0.9193 ± 0.0004 ; log 0.9188 ± 0.0004 ; poly 0.9166 ± 0.0004 ; sine 0.9148 ± 0.0004), the 2022 ordering with spline/log best; the held-out log-likelihood gain of $g+f$ over g -only is $100.5[76.5, 126.9]$ nats (positive in 100% of split $\times$ bootstrap runs). The fitted $\hat{f}$ is concave: the concavity ratio $f(d_{AC})/[f(d_{AB}) + f(d_{BC})]$ for $\Delta L^*=15+15$ segments centered mid-axis is $0.767[0.752, 0.782]$ (spline; log form $0.776[0.763, 0.789]$), matching the value implied by the 2022 report ($2.06/2.72 = 0.757$); the fitted log-form parameters are $a = 3.80[3.29, 4.37]$, $b = 2.37[2.30, 2.44]$ (2022: $a=5.34$, $b=2.34$; b agrees closely, a is the weakly identified shape parameter of the log family and lands in the same regime). The fitted $\hat{g}$ deviates from linearity by at most $8.0[7.3, 8.7]$ L^* units, the crispening bump near the background of the 2022 study, supporting the fixed cube-root position map of Sec. 3.1 as a close stand-in. Within the main model family, the in-family replication gives held-out $M1-M0 = 507.0[309.0, 717.2]$ nats on the Bujack block alone and $74.0[37.1, 114.8]$ nats on the Wiebel block alone—DR reproduces on a suprathreshold scale in both gray datasets, independently. As a further descriptive check of the position map, a standard MLDS scale fit to the Wiebel triads deviates from the cube-root map by at most 0.051 (on a 0–1 scale; correlation 0.9988), versus 0.214 from the linear-in-luminance map—an independent validation of the fixed transducer on a second dataset.

4.4 Disclosures and deviations

(1) The task’s data inventory mentions an $LMS \leftrightarrow XYZ$ transform file; only the two spectral tables were provided, so the bridge matrix is estimated by least squares and verified on known colors (Sec. 3.1), and the $2^\circ/10^\circ$ observer mismatch it spans is disclosed. (2) The cube-root transducer is part of the stated working-space definition, justified empirically by the replication and swept in sensitivity. (3) Wiebel is restricted to the plain context; Helm to color-normal observers. (4) Ekman is placed spectrally (its documentation’s prescription) after the matrix route produced small negative S on the locus. (5) Folds are triad-disjoint for the 2AFC sets (no participant IDs in the release). (6) One correction to the pre-registered text: the plan’s nesting shorthand wrongly claimed $M1 \subset M2c$; $M2c$ ties the weights, so the models are non-nested (comparisons unchanged; discovered by a train-NLL nesting audit and recorded in the attempts log). (7) The CIE RGB generator in the synthetic study is a disclosed post-hoc addition (Sec. 3.5). (8) An initial main-comparison run was discarded wholesale for a compute defect (BLAS thread oversubscription) before any results were read, then

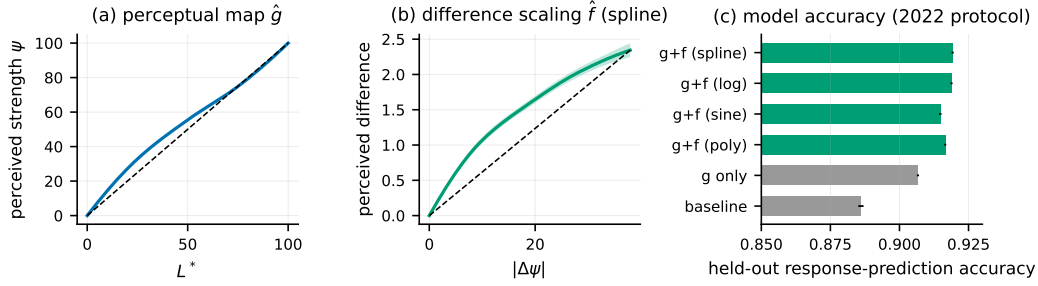


Figure 1: Gray-axis replication of Bujack et al. [2022]. (a) $\hat{g}$ (mean, 95% band over 10 splits $\times$ 100 bootstraps) is near-identity with the 2022 crispening deviation. (b) $\hat{f}$ (spline) is concave—diminishing returns; dashed lines mark additivity. (c) Held-out accuracy by model, 2022 protocol: the 2022 ordering reproduces.

rerun identically. No other deviations occurred: only provided data and fundamentals are used, observation models are per-dataset, the CV discipline is enforced in code, and every specification tried is reported (Sec. 5.5).

5 Experimental Results

5.1 Main comparison

Table 1 is the primary results table; Figure 2 shows every model’s held-out performance. Three findings structure everything that follows.

(i) DR is decisive; its direction-dependence is not. The isotropic-DR model M1 beats the additive baseline M0 by 605.7[395.6, 818.3] held-out nats—diminishing returns replicate emphatically. But per-cone M2 improves on M1 by only 8.2[−4.9, 21.7] nats (interval includes zero), and coarse M2c by 3.8[−21.5, 27.5]: on these public data a single global concave scaling is a strong summary. Per dataset (Table 4, appendix) the only significant M2–M1 gain is on Ekman (5.8[2.2, 9.0] nats)—the S -axis curvature M2 fits (Sec. 5.2) earns its keep only where S varies over a large range.

(ii) The cone orientation loses to standard non-cone bases. Every member of the matched non-cone panel except CIELAB matches or beats M2 in total held-out likelihood (Fig. 2). The best panel member on the full data is the CIE RGB basis; the selection-aware headline delta is

$$\Delta_{\text{heldout}}(\text{M2} - \text{best M3}) = -73.7[-132.4, -31.3] \text{ nats}$$

(negative and significant; per-resample counts of the re-identified best control: opp: 211; srgb: 80; cierrgb: 604; lab: 4; hcl: 101, of $B=1000$). Fixed-pair deltas: M2–opponent = −54.7[−108.1, −7.1]; M2–CIE RGB = −73.7[−127.1, −18.5]; M2–sRGB = −49.3[−108.5, 23.3]; M2–CIE LCh = −25.1[−112.2, 73.0]; M2–CIELAB = 19.3[−84.4, 125.1]. The cone-alignment hypothesis does not merely fail to win—the cone axes are significantly *worse* than two standard non-cone systems. Whether this reversal is trustworthy given the design’s limits is what the synthetic analysis answers (Sec. 5.3): the probability of a control significantly beating M2 this way under cone-aligned truth is 0.00 (zero occurrences in 50 replicates).

(iii) The winning bases share an opponent-like axis. The full-data fits (Table 3, appendix) show why. The CIE RGB basis’ first axis is, in cone coordinates, $u_R = (0.73, -0.68, 0.01)$ —essentially a red–green opponent (L – M -like) direction—and the fit assigns it extreme inner curvature ($\beta = 8.69$, versus ≤ 0.23 for its M -like and S -like axes). The sRGB basis’ first axis is similar ($u = (0.71, -0.70, 0.05)$), and the opponent control carries its weight on the explicit L – M axis. Every basis that outperforms the cones thus contains an L – M -like axis and compresses red–green differences strongly; the cone basis cannot express this, since it applies compression to L and M separately, before they can be differenced.

Table 1: Primary results. Held-out log-likelihoods (nats; higher better): totals over the ten test folds of the joint fit; 95% CIs from the stratified cluster bootstrap over held-out units ($B=1000$); best-control selection re-done inside every resample for the selection-aware row. All numbers generated from the run logs.

Quantity	Estimate	95% CI	Comparison	Notes
Gray-axis DR replication (M1 vs M0, held-out; magnitude vs 2022)	+507.0	[309.0, 717.2]	(2022: concave f ; ratio 0.757)	ratio 0.767 [0.752, 0.782]; Wiebel +74.0 [37.1, 114.8]
Held-out fit, M0 (additive)	-65265.8	[-66255.8, -64267.2]	joint, 4 datasets	2 shape par.
Held-out fit, M1 (isotropic DR)	-64660.1	[-65635.1, -63682.4]	joint	3 shape par.
Held-out fit, M2 (per-cone DR)	-64652.0	[-65609.0, -63687.5]	joint	6 shape par.
Held-out fit, M3 opponent	-64597.2	[-65566.9, -63610.7]	joint	matched complexity
Held-out fit, M3 sRGB	-64602.6	[-65603.8, -63626.5]	joint	matched
Held-out fit, M3 CIE RGB	-64578.3	[-65567.5, -63610.0]	joint	matched
Held-out fit, M3 CIELAB	-64671.3	[-65664.4, -63678.6]	joint	matched
Held-out fit, M3 CIE HCL/LCh	-64626.9	[-65618.7, -63636.1]	joint	matched
M2 vs M1 held-out Δ (anisotropy)	+8.2	[-4.9, 21.7]	sign + significance	positive, n.s.
M2 vs <i>best</i> M3 Δ (cone-alignment, headline)	-73.7	[-132.4, -31.3]	negative, significant	best: CIE RGB (selection-aware)
Identifiability: resolvable granularity + powered channel comparison	$\beta_L - \beta_M$: -0.12[-0.35, 0.02] (n.s.); $\beta_S - \beta_L$: -1.05[-1.79, -0.57] (distinct)		M2c vs M3c-opp: +24.6 [-24.4, 68.7]	granularity: $\{L \approx M\}$ vs $\{S\}$; power Sec. 5.3
Hue-curvature confound (before/after)	before: arc/direct = 2.28 (Ekman), 1.62 (Helm) > 1		after: excess over additive-chord = -0.35 / +0.11[-0.02, 0.26]	anti-cone survives restriction (selection-aware) -64.1 [-138.7, -30.1]

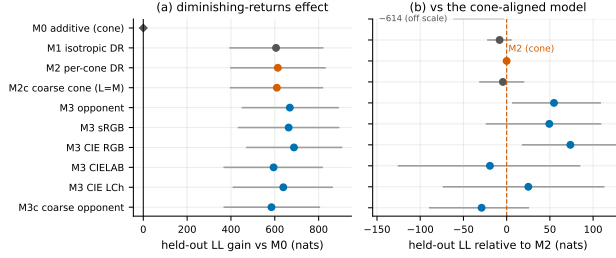


Figure 2: Held-out comparison (all models, joint fit): (a) gain over the additive baseline M0 with 95% cluster-bootstrap intervals; (b) relative to M2. All models share folds, whitening, and complexity class. The DR gain dwarfs differences among DR models; among those, the cone-basis models (M2, M2c) sit at the bottom, below the opponent, sRGB, and CIE RGB controls.

5.2 The fitted per-cone scalings and their distinctness

Within the cone family, M2’s fitted inner scalings are genuinely distinct—but not in the way a three-channel decomposition requires (Fig. 3). The refit-bootstrap intervals are $\beta_L = 0.13[0.05, 0.22]$, $\beta_M = 0.26[0.09, 0.48]$, $\beta_S = 1.18[0.76, 1.90]$ (weights $w_L = 0.66[0.50, 0.84]$, $w_M = 1.65[1.49, 1.80]$, $w_S = 0.69[0.55, 0.82]$; outer $\alpha = 0.42[0.33, 0.52]$). Two different notions of distinctness must be separated. In *shape* (curvature), the S axis is significantly more compressive than either of the others ($\beta_S - \beta_L = -1.05[-1.79, -0.57]$; $\beta_S - \beta_M = -0.93[-1.70, -0.38]$) while L and M are indistinguishable ($\beta_L - \beta_M = -0.12[-0.35, 0.02]$)—so M2 is not a degenerate collapse to M1, and the resolvable shape granularity is the two-class split $\{L \approx M\}$ vs. $\{S\}$. In *weight*, the conditional-on-parametrization estimates do separate ($w_M > w_L$ with non-overlapping intervals), yet tying the channels outright costs almost nothing in prediction: M2c reaches within 4.3[-19.5, 30.8] nats of M2 held-out, and gains only 3.8[-21.5, 27.5] over M1. The weight split is thus a parametrization ridge (with correlated $|\Delta \tilde{L}|$, $|\Delta \tilde{M}|$ regressors, other parameters absorb a tie), not predictive structure—exactly what the identifiability analysis below polices. The coarse pre-combination model’s advantage over the matched coarse *post*-combination control is 24.6[-24.4, 68.7] (M2c vs M3c-opp; positive on Ekman 19.3[1.1, 34.3] and Bujack 3.9[0.1, 8.0], absent on Helm 1.0[-44.5, 42.8]).

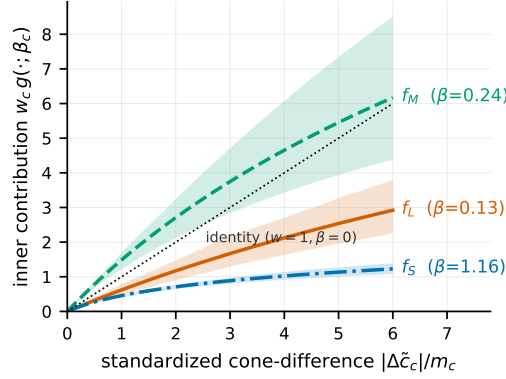


Figure 3: Fitted per-cone channel scalings of M2, $f_c = w_c g(x; \beta_c)$, with 95% refit-bootstrap bands ($B=200$). The S -axis curvature is strongest and significantly distinct; L and M curvatures are indistinguishable (weights differ, but tying the channels costs almost nothing held-out; see text)—resolvable within-cone granularity $\{L \approx M\}$ vs $\{S\}$.

5.3 Identifiability: what these designs can and cannot resolve

Design diagnostics. Pooled over all likelihood pairs, $\text{corr}(|\Delta \tilde{L}|, |\Delta \tilde{M}|) = 0.54$ (Ekman 0.31, Helm 0.41; the gray sets are rank-one: exactly collinear triads, $\leq 1.5e - 06^\circ$). Sampled unit directions have relative singular values (1.00, 0.45, 0.34); only 9.4% of direction variance lies along $L-M$ (vs. 38.6% along $L+M$, 52.0% along S)—precisely where the cone and opponent bases differ. Distinguishing cone- from opponent-aligned compression is therefore a priori hard, which is why the discrimination analysis below must precede any reading of Sec. 5.1(ii).

Synthetic recovery and discrimination power (all synthetic, labeled). Simulating from the fitted M2 at the exact real designs and refitting (50 replicates), the S -axis curvature recovers accurately (β_S : true 1.16, recovered median and 95% quantiles 1.14[0.24, 2.50]), while the L and M curvatures recover only loosely, with wide spread and upward median bias (β_L : true 0.13, recovered 0.27[0.00, 5.43]; β_M : 0.24 $\rightarrow$ 0.28[0.00, 1.13])—direct confirmation that the finest per-cone split is weakly identified at these designs, exactly as the collinearity diagnostics predicted. Discrimination rates across 50 replicates per generator (Fig. 4c) then calibrate every verdict in Sec. 5.1: (i) when the world is truly cone-aligned (M2 generator), the probability that a control *significantly beats* M2—the outcome we actually observed—is 0.00 against the opponent control, 0.00 against CIE RGB, and 0.00 against the selected-best control: zero occurrences in 50 replicates for each. (ii) When a control is the truth, the same comparisons detect it essentially always (0.98 opponent, 1.00 CIE RGB; false-positive rate of M2 beating the opponent control under opponent truth, 0.00). The observed cone deficit is therefore of a kind that does not arise under cone truth and that the design detects reliably when real: the negative headline is *earned*, not an underpowered accident. (iii) The power to certify alignment positively (M2 significantly beating the best of the controls under cone truth) is 0.54 (0.68 against the opponent alone)—so a true cone alignment would have been certified only about half the time; the test is markedly better at detecting misalignment than at proving alignment. (iv) Anisotropy is underpowered: M2-over-M1 has power 0.34 under cone truth (false-positive rate 0.00), so the non-significant M2–M1 delta on the real data is uninformative about direction-dependence—absence of evidence, not evidence of absence. (v) At the coarse level the pre- vs. post-combination comparison is well powered where it matters: M2c beats the matched non-cone coarse control in 1.00 of replicates when coarse-cone structure is true (false-positive rate under the opponent generator 0.00), so its real-data null (Table 1) is informative evidence *against* coarse-cone organization at the fitted effect size; M2c-over-M1 has power only 0.16, so its null carries little weight.

5.4 Hue-curvature confound: before and after

Before control, both hue circles look strongly subadditive: arc sums exceed direct dissimilarities by factors of 2.28 (Ekman) and 1.62 (Helm) on average—naively, chromatic DR everywhere. *After control*, most of this is geometry (plus, for Ekman, response saturation): the fitted *additive*

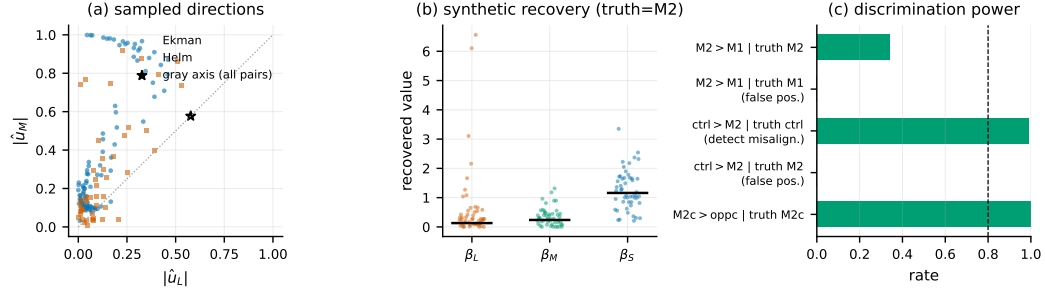


Figure 4: Identifiability. (a) Sampled pair directions (L, M components): the gray sets collapse to one achromatic direction (star); the hue circles undersample the $\Delta L \approx \Delta M$ subspace. (b) Synthetic recovery of $(\beta_L, \beta_M, \beta_S)$ under the M2 generator (black bars: truth). (c) Discrimination rates (SYNTHETIC): powers (green), false positives (vermilion); dashed line 0.8.

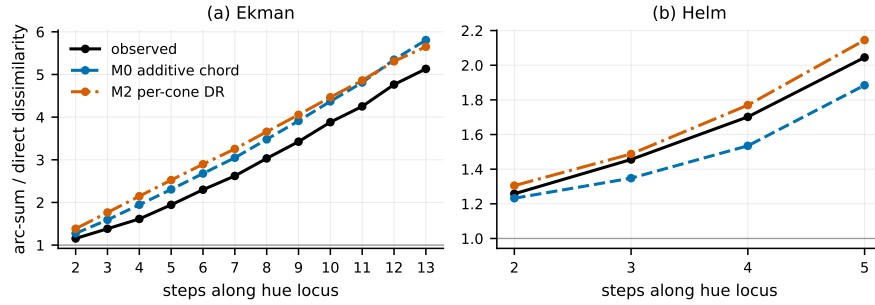


Figure 5: Curvature control on the hue circles: mean arc-sum/direct ratio by locus separation—observed vs. the additive-chord (M0, no-DR) baseline and the M2 prediction, through the fitted links. Most observed subadditivity is reproduced by the additive model’s chord geometry (plus, for Ekman, link saturation); the rating-data DR verdict rests on the model comparison, not raw subadditivity.

chord model M0, with no DR at all, already predicts ratios 2.62 and 1.50 through the same fitted links (Fig. 5). The observed excess over the additive-chord account is -0.35 for Ekman (negative: geometry plus the saturating link already over-predict the observed subadditivity, leaving nothing for DR to explain on that response scale) and $+0.11[-0.02, 0.26]$ for Helm (observer-level bootstrap; positive-leaning but not significant—at most a hint of chromatic DR on the one rating set with a linear link). Chromatic DR *presence* on the rating datasets is thus not separately established here beyond the link/curvature account—consistent with the pre-registered position that outer-DR evidence comes from the forced-choice data—while the *alignment* comparisons, which are link-invariant, are unaffected. Restricting to near-adjacent pairs (locus close to straight; 36 of 91 Ekman and 220 of 495 Helm rows retained), every headline conclusion survives: $M1-M0 = 593.6[406.9, 801.8]$; $M2-M1 = 2.6[-6.3, 11.4]$ (n.s.); the cone basis still loses significantly ($M2-opp = -64.1[-137.4, -5.3]$, $M2-sRGB = -54.7[-103.3, -9.1]$, $M2-CIE RGB = -36.5[-67.0, -7.0]$; selection-aware $-64.1[-138.7, -30.1]$, best control now opponent). The anti-cone verdict is not a curvature artifact.

5.5 Specification sensitivity (all specifications reported)

Four pre-registered variants re-run the entire pipeline: transducer exponent 0.43 (IPT-like) instead of $1/3$; linear instead of saturating Ekman link; saturating instead of linear Helm link; and a power-family inner scaling $g(x; \gamma) = x^\gamma$ instead of the log family. Table 2 reports the key deltas under every variant; no specification was selected on its outcome, and the primary was fixed in the plan. *First*, in no specification does the cone basis match the best control: the selection-aware delta is negative in all five (significantly in the primary, 0.43-exponent, and linear-Ekman variants; marginally spanning zero in the other two). *Second*, the anisotropy verdict is link-dependent as expected: with the linear Ekman link—which cannot absorb curvature into the response scale— $M2-M1$ becomes

Table 2: Sensitivity sweep: key held-out deltas (nats, 95% cluster-bootstrap CIs; selection-aware for the best-control column) under every specification tried.

Specification	M2 – M1	M2 – best M3 (sel.-aware)	best control
primary (cube root; Ekman Shepard; Helm linear; log inner)	8.2[–4.9, 21.7]	–73.7[–132.4, –31.3]	ciergb
transducer exponent 0.43	11.4[–4.5, 27.2]	–755.7[–928.0, –588.9]	hcl
Ekman link linear	24.7[11.0, 38.2]	–75.2[–141.8, –41.1]	ciergb
Helm link saturating	3.8[–4.3, 12.7]	–35.1[–98.3, 3.6]	ciergb
inner family: power	43.3[–20.1, 115.4]	–38.2[–114.3, 11.9]	ciergb

significantly positive (24.7[11.0, 38.2]), the S -axis compression then carried by the metric itself; under the conservative primary link it is absorbed and M2–M1 is not significant. *Third*, the extreme deficit under exponent 0.43 (–755.7[–928.0, –588.9], best control CIE LCh) is a transducer effect, not an alignment effect: it is dominated by the achromatic blocks, where the 0.43-power map misfits the gray data that the cube-root map (and CIELAB’s own L^* , untouched by this sweep) fits well—independent confirmation of the cube-root convention, with the chromatic anti-cone margins persisting (CIE LCh still beats M2 on Helm under that variant).

5.6 Supplement: free-orientation analysis

Fitting the same six-parameter family on a *fitted* rotation of the cone axes (three extra parameters; nests M2 and the linear controls; orientation recovery, not a matched comparison) asks where the data put the axes when free. The best-fitting axis set sits 44.4° from the cone axes (worst-matched axis, permutation- and sign-invariant) and 42.5° from the opponent axes; across refit bootstraps the distance-to-cones is 43.5[21.5, 52.1]° and the distance-to-opponent 44.3[33.3, 66.0]°. The supplement confirms the *negative* cleanly—the confidence region of the free axes excludes the cone orientation—while not certifying the opponent set either: the leading free axis is L – M -like (the panel’s motif), but the remaining axes rotate away from every named system and one runs to the curvature bound—a reminder that this exploratory fit, with three extra parameters, exploits structure (and noise) the fixed-basis comparison deliberately excludes. Indeed, held out, the free-orientation model reaches –64607.9 nats—better than M2 (–64652.0) but *below* the best fixed controls (CIE RGB –64578.3, opponent –64597.2): the three extra rotation parameters do not pay for themselves out of sample, which is precisely why the headline test uses fixed bases.

6 Analysis

Verdicts. (a) **Is DR direction-dependent at all (M2 vs M1)?** Weakly, on these data: the point estimate is positive (8.2[–4.9, 21.7] nats) but the interval includes zero; only the Ekman block gains significantly (5.8[2.2, 9.0]), where the huge S range lets the compressive S scaling matter. Since Stark et al. [2025] demonstrated non-constancy with purpose-built chromatic-line data, the right reading is not that DR is isotropic but that *public* data span too few directions, with too little per-direction power, to demonstrate anisotropy—an even stronger form of the data limitation their “cannot be tested” remark identified. (b) **Is the structure cone-aligned (M2 vs best M3)?** No—and the evidence is stronger than a mere failure to win. The cone basis is significantly outperformed by the CIE RGB and opponent bases (selection-aware deficit –73.7[–132.4, –31.3] nats), the deficit survives the curvature restriction (–64.1[–138.7, –30.1]), remains negative under every specification tried (significantly in three of five; Table 2), and lies in a direction that did not arise in any of 50 cone-truth replicates (0.00 under the M2 generator). In no analysis, under no specification, does the cone-aligned model match the best standard non-cone basis: the per-cone decomposition of Eq. (1), as a privileged description, is rejected on these designs. (c) **At what granularity is channel structure identifiable?** Within the cone family, a two-class split only— β_S distinct ($\beta_S - \beta_L = -1.05[-1.79, -0.57]$), L vs M unresolved ($\beta_L - \beta_M = -0.12[-0.35, 0.02]$), as the collinearity diagnostics predicted (9.4% of direction variance along L – M). Across families the data *can* distinguish cone- from opponent/CIE-RGB-aligned compression (rates above), and they choose the latter. The coarse pre- vs post-combination comparison (M2c vs M3c-opp) is positive for pre-combination on Ekman and Bujack but null overall (24.6[–24.4, 68.7]), so the site question is answered only at the fine level, where the post-combination (opponent-like) organization wins.

Mechanistic reading (calibrated). Three independent fits—the CIE RGB control (its $u_R = (0.73, -0.68, 0.01)$ axis given $\beta = 8.69$), the sRGB control, and the opponent control—converge on the same motif: strong compression along a red–green opponent-like axis. Compression applied to an L – M -like signal is, by construction, *post-receptoral*: it operates on a quantity that exists only after the cone signals are differenced. The natural (not uniquely mandated) picture is strong difference-scaling nonlinearity in the red–green opponent channel, plus the achromatic DR of Bujack et al. [2022] and a compressive S -axis contribution visible in the Ekman geometry. We flag the limits of this reading explicitly: the “best basis” (CIE RGB) is one member of a panel of similar opponent-containing bases and its margin over the opponent basis itself is not the object of a controlled claim; and the Helm dataset contributes the largest share of the anti-cone margin (M2–CIE RGB per dataset: Bujack 5.4[0.2, 10.5], Wiebel $-0.5[-1.9, 1.0]$, Ekman 2.3[$-8.8, 11.0$], Helm $-80.9[-130.5, -29.6]$), so the verdict leans on the surface-color hue circle; the restricted-pair reanalysis and the Ekman agreement mitigate but do not eliminate this concentration.

Failure analysis (what could not be established, and why). (1) *Three-way per-cone granularity*: unidentifiable here; L/M separation would need stimuli that decouple $|\Delta L|$ from $|\Delta M|$ (directions near $\Delta L \approx \Delta M$ off the achromatic ray), which no public dissimilarity set provides. (2) *Positive certification of alignment* (had it been true): power 0.54 against the best of the controls—public data would have certified a true cone alignment only about half the time, while detecting misalignment nearly always (0.98–1.00); the design refutes far better than it proves. (3) *Outer-DR on rating data*: confounded with the response link by construction; our DR-presence claims rest on the forced-choice sets only. (4) *Cross-dataset curvature transfer*: gray data pin achromatic DR, hue circles pin chromatic structure, and their link is conventional (shared functional form, von-Kries normalization); a single dataset spanning both direction families does not exist publicly. (5) *Boundary/nuisance degeneracies*: some control fits park weight at zero on an axis (the opponent fit sets $w_{L+M} \rightarrow 0$, routing achromatic content through its $S-(L+M)$ axis, which overlaps the achromatic direction); where $w_k \approx 0$ the paired β_k is unidentified—affecting control-parameter interpretation, not the held-out comparisons. (6) *Aggregation family*: all models aggregate additively (L1) across axes before the outer scaling; an L2 (Riemannian-like) aggregation variant is outside the pre-registered family, so the verdicts concern the nonlinearity’s orientation within additive aggregation. (7) The M2c-vs-M1 comparison is non-nested (weights tied vs free), discovered in an audit and disclosed; conclusions do not rest on it.

7 Conclusion

We set out to test the hypothesis Stark et al. [2025] posed but could not test: that diminishing returns decomposes along the cone axes (Eq. 1). Assembling four public datasets in a normalized CIE 2006 cone space, anchored by a full replication of the 2022 gray-axis DR finding (held-out gain 507.0 nats; concavity ratio 0.767[0.752, 0.782] vs their 0.757), our pre-registered, selection-aware, identifiability-audited comparison finds: diminishing returns replicates decisively; its direction-dependence is weak on public data; and where per-axis structure is resolvable it is *not* cone-aligned—the cone basis is significantly outperformed by standard bases containing a red–green opponent-like axis, in a comparison whose error rates we measured, which survives curvature control, and which stays negative under every specification swept. The compact per-cone generative model of DR is therefore not supported: if a compact description exists, our evidence points to compression organized along post-receptoral, opponent-like axes—compression *after* the cone signals are combined—with the S -dominated compression on the saturated hue circle as the one cone-flavored structure the data do resolve. Beyond the specific verdict, this study contributes what the conjecture lacked: a quantitative statement of what public dissimilarity data can and cannot decide about channel structure (three-way cone splits: no; detecting a misaligned cone frame: yes, nearly always; positively certifying alignment: only with coin-flip power), and a reusable, fully logged harness for such tests. The decisive experiment public data cannot deliver is clear: suprathreshold dissimilarities along directions decoupling ΔL from ΔM —intermediate between the achromatic ray and the isoluminant plane—ideally with the triad methodology of Bujack et al. [2022], whose forced-choice design pins the scaling that rating data cannot.

Limitations. The verdicts are conditional on the stated working space (von-Kries cube-root cone coordinates; exponent swept), on additive cross-axis aggregation, on the fixed-basis construction of

the controls, and on datasets collected under different viewing conditions harmonized only through per-dataset whites and links; the anti-cone margin is carried disproportionately by the Helm tiles; and the mechanism reading (post-combination compression) is an interpretation of the winning bases' shared structure, not a directly manipulated variable.

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A Provenance and licensing

All code, seeds, fold assignments, fit logs, and figure/table generators are under proposal/; every number above is typeset from machine-generated macros derived from those logs; the pre-registration is proposal/analysis_plan.md and deviations are catalogued in Sec. 4.4 and proposal/attempts_log.md. Dataset licenses: lanl/color BSD-3; Wiebel et al. CC-BY 4.0; Ekman/Helm matrices via smacof (GPL-3); cone fundamentals from cvrl.org under its citation terms.

B Cone-conversion verification detail

The least-squares bridge M ($XYZ \rightarrow LMS$ over the common 390–830 nm grid) achieves per-cone $R^2 = 0.9992/0.9992/0.9781$; equal-energy-white check: matrix vs exact spectral LMS agree to 0.46° (worst component 2.9%); 600 nm locus check: 0.53° . Neutral stimuli map to exactly (t, t, t) after von Kries for any white (verified). The matrix route is weakest for near-zero- S monochromatic lights (negative S at 537, 555 nm), which is why Ekman uses direct spectral placement; the two routes agree to $\leq 17.8^\circ$ in working-space direction across the 14 Ekman stimuli, with the largest deviations at the near-zero- S wavelengths. Whitening constants m_k and all per-stimulus coordinates are logged.

C Full-data fitted parameters and information criteria

Table 3 lists every model’s full-data parameters and AIC (the AIC/BIC ordering matches the held-out ordering; both are reported for transparency only—no claim rests on in-sample criteria). Where $w_k \approx 0$ the paired β_k is unidentified (Sec. 6).

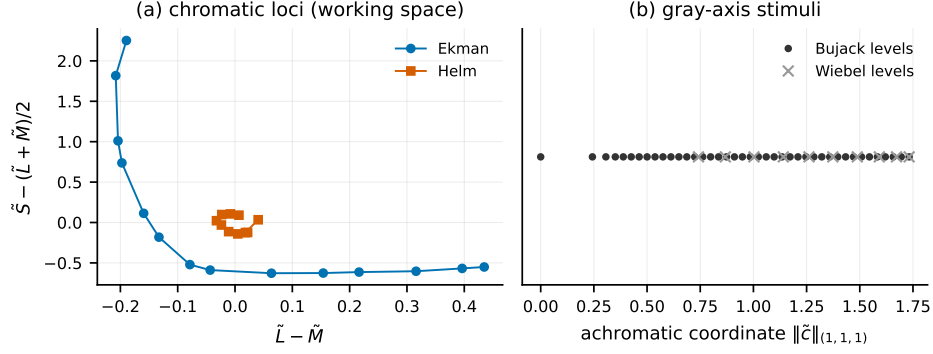


Figure 6: Stimulus geometry in the working cone space $\tilde{c}$ (cube-root von-Kries CIE 2006 coordinates). (a) The two chromatic loci in an opponent-plane projection: Ekman’s 14 monochromatic lights (saturated, spanning a wide loop) and Helm’s 10 Munsell tiles (moderate chroma). (b) The gray-axis stimulus levels of the two achromatic datasets along the neutral ray. The gray sets sample a single direction (exactly collinear triads); the hue circles sample many chromatic directions; directions with $\Delta\tilde{L} \approx \Delta\tilde{M}$ off the achromatic ray—one of the two direction families on which the cone and opponent bases disagree most—are sparsely covered (Sec. 5.3).

Table 3: Full-data fits (joint, all four datasets): per-axis weights w , inner curvatures β (axis order: cone L, M, S ; opponent $L+M, L-M, S-(L+M)$; RGB bases R, G, B ; CIELAB L^*, a^*, b^* ; LCh L^*, C^*, H^*), outer α , in-sample NLL, AIC.

Model	w (per axis)	β (per axis)	α	in-sample NLL	AIC
M0	(1.08, 1.51, 0.41)	(0.00, 0.00, 0.00)	0.00	65193.0	130422.0
M1	(0.87, 1.71, 0.43)	(0.00, 0.00, 0.00)	0.57	64582.8	129203.6
M2	(0.66, 1.65, 0.69)	(0.13, 0.24, 1.16)	0.42	64572.9	129189.7
M2c	(1.21, 1.21, 0.58)	(0.17, 0.17, 0.69)	0.37	64583.9	129207.7
M3-opp	(0.00, 2.06, 0.93)	(0.00, 0.00, 0.08)	2.23	64529.2	129102.4
M3-srgb	(0.72, 0.63, 1.65)	(0.00, 0.01, 0.06)	0.77	64512.8	129069.6
M3-ciergb	(0.80, 1.60, 0.59)	(8.69, 0.23, 0.21)	0.48	64502.0	129048.0
M3-lab	(0.09, 1.88, 1.03)	(0.45, 0.00, 0.00)	7.02	64593.6	129231.3
M3-hcl	(0.12, 0.01, 2.87)	(0.43, 300.00, 0.45)	5.34	64543.8	129131.6
M3c-opp	(1.25, 0.87, 0.87)	(0.21, 1.99, 1.99)	0.48	64581.2	129202.5

D Per-dataset held-out deltas

E Across-fold variation

Fold-level deltas for the key comparisons (each fold’s test log-likelihood difference under that fold’s training fit): $M1 - M0 = 60.6 \pm 29.7$ (10/10 folds > 0); $M2 - M1 = 0.8 \pm 1.5$ (7/10 folds > 0); $M2 - \text{CIE RGB} = -7.4 \pm 7.8$ (2/10 folds > 0); $M2 - \text{opponent} = -5.5 \pm 7.2$ (2/10 folds > 0); $M2c - M3c\text{-opp} = 2.5 \pm 8.8$ (6/10 folds > 0). Table 5 lists every fold.

F Reproducibility

Base seed 20260816; every stochastic step derives a logged child seed (CRC-based, process-independent). Pipeline: `run_convert.py` (conversion + verification) $\rightarrow$ `run_design_diagnostics.py` $\rightarrow$ `run_gray_replication.py` $\rightarrow$ `run_main.py` [tag] $\rightarrow$ `run_deltas.py` [tag] $\rightarrow$ `run_confound.py`, `run_boot_params.py`, `run_synth.py`, `run_rotation.py` $\rightarrow$ `make_figures.py`, `make_macros.py`. Fits: L-BFGS-B, 12 multistarts (6 in synthetic study), convergence and per-start objectives logged; 1.000 of cross-validated best fits report clean convergence, and the train-NLL nesting audit ($M0 \supset M1 \supset M2$ within numerical tolerance on every fold) is in `logs/nesting_checks.json`. Total compute: a few CPU-hours on 32 cores; peak memory far below limits.

Table 4: Per-dataset decomposition of key held-out deltas (nats; 95% cluster-bootstrap CIs). Positive favors the first model.

Δ	Bujack	Wiebel	Ekman	Helm
M1 – M0	506.2[304.9, 717.6]	75.1[35.8, 118.9]	-4.1[-10.2, 3.3]	28.5[9.0, 48.2]
M2 – M1	3.3[-1.9, 8.3]	-0.6[-3.2, 2.1]	5.8[2.2, 9.0]	-0.3[-12.3, 12.0]
M2 – M3-opp	0.3[-3.3, 3.5]	-0.9[-2.5, 0.9]	-3.9[-21.3, 9.9]	-50.2[-102.0, -4.5]
M2 – M3-CIE RGB	5.4[0.2, 10.5]	-0.5[-1.9, 1.0]	2.3[-8.8, 11.0]	-80.9[-130.5, -29.6]
M2 – M3-sRGB	0.8[-2.7, 3.9]	-0.9[-2.7, 1.0]	3.0[-6.9, 9.6]	-52.3[-110.5, 20.4]
M2 – M3-CIELAB	62.2[17.4, 115.0]	-0.4[-1.0, 0.2]	29.8[16.3, 45.3]	-72.3[-160.2, 12.5]
M2 – M3-LCh	62.3[17.4, 115.0]	-0.4[-1.0, 0.2]	2.7[-23.2, 24.6]	-89.6[-157.1, -14.4]
M2c – M3c-opp	3.9[0.1, 8.0]	0.4[-0.2, 0.9]	19.3[1.1, 34.3]	1.0[-44.5, 42.8]

Table 5: Per-fold held-out deltas (nats), fold f0–f9 of the fixed 10-fold split.

Δ (nats)	f0	f1	f2	f3	f4	f5	f6	f7	f8	f9	mean $\pm$ SD
M1–M0	81.5	35.2	30.0	92.3	80.2	81.6	54.0	13.2	40.2	97.5	60.6 $\pm$ 29.7
M2–M1	2.3	-2.2	-0.0	-0.9	2.1	0.4	0.9	1.6	2.2	1.9	0.8 $\pm$ 1.5
M2–CIE RGB	-20.0	-11.5	-4.1	-13.7	-5.4	0.8	-8.4	-6.9	-12.1	7.5	-7.4 $\pm$ 7.8
M2–opp	-7.0	-1.2	-4.3	-23.7	1.6	-2.1	-8.4	-4.9	-5.9	1.1	-5.5 $\pm$ 7.2
M2c–M3c	8.5	12.1	3.0	-12.8	9.5	-3.8	-4.2	10.3	-7.1	9.2	2.5 $\pm$ 8.8